AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

AI-Powered Student Feedback Intelligence: Transforming Insights into Teaching Excellence”

Educational Analytics & NLP

Excel + Power BI + ChatGPT

A Project Report

submitted in partial fulfillment of the requirements of the AICW Project

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Year - 2026

ACKNOWLEDGEMENT

I would like to take this opportunity to express my deep sense of gratitude to all individuals who helped me directly or indirectly during this project work.

Firstly, I would like to thank our supervisor, Mr. Ankit, for being a great mentor and the best adviser I could have asked for. His advice, encouragement, and constructive feedback served as a source of innovative ideas and inspiration, and were central to the successful completion of this project. The confidence shown in me was the biggest source of motivation throughout this journey.

I extend my sincere thanks to the Edunet Foundation, Microsoft, LinkedIn, and SAP for providing this fellowship opportunity, the learning resources, and the platform to develop and showcase my AI skills.

I am also grateful to our institution, Shri L R Tiwari Degree College, for providing the necessary infrastructure and support throughout the project duration.

Finally, I thank my family and friends for their continuous encouragement and support.

ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

This project addresses the challenge of extracting meaningful insights from large volumes of student feedback, which often remain underutilized due to inconsistencies, unstructured formats, and subjective language. The primary objective was to design an AI-powered system that transforms raw feedback into actionable intelligence for educators, enabling continuous improvement in teaching practices. Leveraging natural language processing (NLP) techniques, sentiment analysis, and visualization tools, the methodology involved cleaning and structuring feedback data, categorizing comments into themes, and identifying sentiment trends. Microsoft Copilot and ChatGPT was integrated to assist in synthesizing insights and building interactive dashboards that highlight key performance indicators such as student satisfaction, teaching effectiveness, and areas requiring attention. Results demonstrated that the system successfully reduced noise in feedback data, uncovered hidden patterns, and provided clear, data-driven recommendations for educators. The dashboards enabled real-time monitoring of student perceptions, while AI-assisted analysis offered deeper contextual understanding of recurring issues. Overall, the project concludes that AI-driven feedback intelligence can significantly enhance teaching quality by bridging the gap between student experiences and instructional improvement, ultimately fostering a more responsive and effective learning environment.

Keywords: AICW, LinkedIn, Feedback, Effectiveness, Support

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CHAPTER 1

This chapter introduces the project undertaken as part of the **AICW Fellowship Program**. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting. The chosen project, AI-Powered Student Feedback Intelligence: Transforming Insights into Teaching Excellence, focuses on leveraging artificial intelligence to analyze and interpret student feedback data, thereby enhancing teaching quality and learning outcomes.Student feedback has long been recognized as a vital mechanism for improving teaching quality and enhancing the overall learning experience. Universities and schools routinely collect feedback through surveys, course evaluations, and open-ended comments. However, the sheer volume of responses, coupled with their unstructured and subjective nature, often makes it difficult for educators to derive meaningful insights. Traditional methods of analyzing feedback—such as manual review or basic statistical summaries—are time-consuming, prone to bias, and frequently fail to capture the nuanced perspectives of students. As a result, valuable information that could guide instructional improvement remains underutilized.

In the era of digital transformation, educational institutions are increasingly exploring data-driven approaches to decision-making. Artificial Intelligence (AI), particularly Natural Language Processing (NLP) and sentiment analysis, offers powerful tools to address the challenges of feedback interpretation. By automating the cleaning, categorization, and visualization of student responses, AI can transform raw data into actionable intelligence. This project builds on that potential, aiming to bridge the gap between student experiences and teaching excellence.

1.1 Problem Statement

The core problem addressed in this project is the under utilization of student feedback in educational institutions. While feedback is routinely collected through surveys, course evaluations, and open-ended comments, it often remains unanalyzed or is processed using basic statistical methods. This leads to several challenges:

- **Unstructured Data:** Student feedback is largely qualitative, expressed in free-text form with varied language, spelling errors, and subjective phrasing. Traditional tools struggle to process such data effectively.
- **Volume of Responses:** Large institutions generate thousands of feedback entries, making manual review impractical and time-consuming.
- **Noise and Bias:** Feedback often contains irrelevant comments, emotional bias, or contradictory statements, which obscure meaningful insights.
- **Delayed Action:** By the time feedback is processed, opportunities for timely intervention may already be lost, reducing its impact on teaching improvement.

This problem is significant because it directly affects teaching quality, student satisfaction, and institutional reputation. Educators, administrators, and students are all impacted: educators lack actionable insights, administrators cannot make evidence-based decisions, and students feel their voices are not adequately heard. Existing approaches—manual reviews or simple survey averages—fail to capture nuanced perspectives and do not provide dynamic, actionable intelligence.

1.2 Motivation

The motivation for undertaking this project stems from both real-world relevan**ce** and professional value. Student feedback is one of the most important mechanisms for evaluating teaching effectiveness, yet it is often underutilized due to its unstructured nature and the limitations of traditional analysis methods. By applying artificial intelligence to this domain, the project seeks to bridge the gap between raw feedback and actionable insights, thereby contributing to teaching excellence.

Personal and Professional Significance

This project is personally meaningful because it aligns with the growing emphasis on student-centered learning and the need for educators to be responsive to student voices. Professionally, it provides valuable experience in applying AI tools such as Microsoft Copilot and ChatGPT, Natural Language Processing (NLP), and sentiment analysis to solve practical problems.

The fellowship context further enhances its importance, as it demonstrates the ability to apply advanced analytics in a structured, outcome-driven environment.

Real-World Applications

The potential applications of this project extend across educational institutions:

- **Course Improvement:** Educators can identify recurring issues in teaching methods, course content, or assessment practices.
- **Quality Assurance:** Administrators can use insights to meet accreditation requirements and demonstrate continuous improvement.
- **Student Engagement:** Students benefit from a more responsive learning environment where their feedback directly influences teaching strategies.
- **Policy Development:** Institutions can design policies based on evidence rather than anecdotal observations.

Organizational and Domain Impact

The impact of this project on the education domain is significant. By transforming feedback into structured intelligence, institutions can enhance teaching quality, improve student satisfaction, and strengthen their reputation. The project also highlights the role of AI in supporting evidence-based decision-making, ensuring that feedback is not just collected but actively used to drive change.

Alignment with Industry Trends

This project aligns closely with modern industry trends in **AI and data analytics**:

- **AI-Powered Insights:** Leveraging NLP and sentiment analysis reflects the growing use of AI to interpret unstructured data.
- **Data-Driven Decision-Making:** The project embodies the shift toward evidence-based practices in education and beyond.
- **Interactive Visualization:** Dashboards and Copilot-assisted reporting mirror industry standards for business intelligence and analytics.

- **Scalability:** The framework demonstrates how AI solutions can be scaled across domains, from education to healthcare and business. In summary, the motivation behind this project lies in its ability to combine personal interest, professional growth, and real-world impact. By aligning with industry trends and addressing a critical challenge in education, the project showcases the transformative potential of AI-powered feedback intelligence.

1.3 Objectives

1. To collect and understand the structure of the dataset used in the project.
2. To clean and preprocess raw data by removing duplicates, correcting formats, and handling missing values.
3. To apply domain-specific analysis using Excel, Power BI, Copilot and ChatGPT.
4. To build interactive dashboards or reports that clearly communicate insights
5. To leverage AI tools (Copilot / ChatGPT) to automate or enhance the analysis and content generation process.
6. To apply Natural Language Processing (NLP) and sentiment analysis for deeper feedback interpretation.
7. To provide actionable recommendations for educators based on AI-driven insights

1.4 Scope of the Project

The scope of this project defines its boundaries, limitations, and areas of application. It ensures clarity on what the project covers, the tools used, and what is intentionally excluded, while also highlighting its role as part of the AICW Fellowship Program.

Dataset Coverage: This project focuses on student feedback data collected through surveys, course evaluations, and open-ended comments. Both qualitative (textual responses) and quantitative (numerical ratings) data are included to provide a comprehensive view of student perceptions.

Tools and Technologies: The tools used are limited to Microsoft Excel, Power BI, and Copilot/AI ChatGPT as assigned under the fellowship program. Excel is employed for preliminary data exploration and cleaning, Power BI for advanced visualization and dashboard creation, and Copilot/AI ChatGPT for AI-assisted analysis, sentiment interpretation, and automated reporting.

Focus Area of Analysis: The analysis emphasizes data cleaning, visualization, and AI-assisted reporting. Specifically, the project applies Natural Language Processing (NLP) and sentiment analysis to categorize feedback into themes and identify sentiment polarity. The primary focus is on transforming unstructured feedback into structured, actionable insights that educators and administrators can use to improve teaching practices.

Out of Scope: The project does not include:

- ✔ Real-time data integration from live feedback systems.
- ✔ Predictive machine learning models for forecasting student performance.
- ✔ Production-level deployment or integration with institutional learning management systems (LMS).
- ✔ Broader institutional analytics such as financial, enrollment, or administrative data.

Intended Outcome: The project is intended as a fellowship showcase rather than a production-level system. Its purpose is to demonstrate the potential of AI-powered feedback intelligence in education, highlighting how modern tools can transform raw student feedback into meaningful insights. While limited in scope, the framework provides a proof of concept that can be scaled and extended in future institutional applications.

CHAPTER 2
Literature Survey

AI-driven insights can suggest areas for improvement, enabling educators to deliver targeted support and interventions where necessary **(Seo et al., 2021)**; uncover student learning behavior patterns, helping teachers to redesign curriculum and instructional strategies **(Hashim et al., 2022; Luan et al., 2020)**; and reveal trends in classroom dynamics and instructional effectiveness **(Wang, 2025; Zhang & Zhang, 2024)**.

2.1 Existing Work and Research

Artificial Intelligence (AI) has emerged as a transformative force in education, particularly in the analysis of student feedback. Traditional feedback evaluation methods, such as manual reviews or basic statistical summaries, are limited in their ability to process large volumes of unstructured data. AI-powered techniques, including Natural Language Processing (NLP) and sentiment analysis, offer new possibilities for extracting meaningful insights from qualitative feedback (Hashim, Omar, Jalil, & Sharef, 2022).

2.2 Tools and Technologies Review

1. Microsoft Excel

Microsoft Excel is one of the most widely used tools for data cleaning, transformation, and basic visualization. It provides functionalities such as pivot tables, conditional formatting, and data validation, which are essential for organizing and preprocessing datasets. In the context of this project, Excel is used to remove duplicates, correct formatting errors, and handle missing values in student feedback data before advanced analysis.

2. Microsoft Power BI

Power BI is a business analytics tool that enables interactive visualizations and business intelligence reporting. Its user-friendly interface allows educators and administrators to create dashboards that highlight trends in student satisfaction, teaching effectiveness, and areas requiring improvement. Power BI's ability to integrate multiple data sources makes it particularly valuable for institutional feedback analysis.

3. Microsoft Copilot (AI)

Microsoft Copilot is an AI assistant integrated into Microsoft 365 tools. It supports automation of tasks, generation of summaries, and assistance with data analysis through natural language prompts. In this project, Copilot is leveraged to synthesize insights from unstructured feedback, automate reporting, and provide contextual recommendations for educators.

4. AI Chatbots (e.g., ChatGPT)

Generative AI chatbots are increasingly used for content creation, marketing copy, report generation, and prompt-driven analysis. In this project, they complement ChatGPT by assisting in thematic clustering of feedback, sentiment interpretation, and generating narrative summaries of student perceptions.

5. Natural Language Processing (NLP) and Sentiment Analysis Tools

Beyond the standard productivity tools, NLP libraries and sentiment analysis frameworks are employed to process unstructured text feedback. These techniques enable categorization of comments into themes (e.g., teaching methods, course content, assessment practices) and identification of sentiment polarity (positive, negative, neutral).

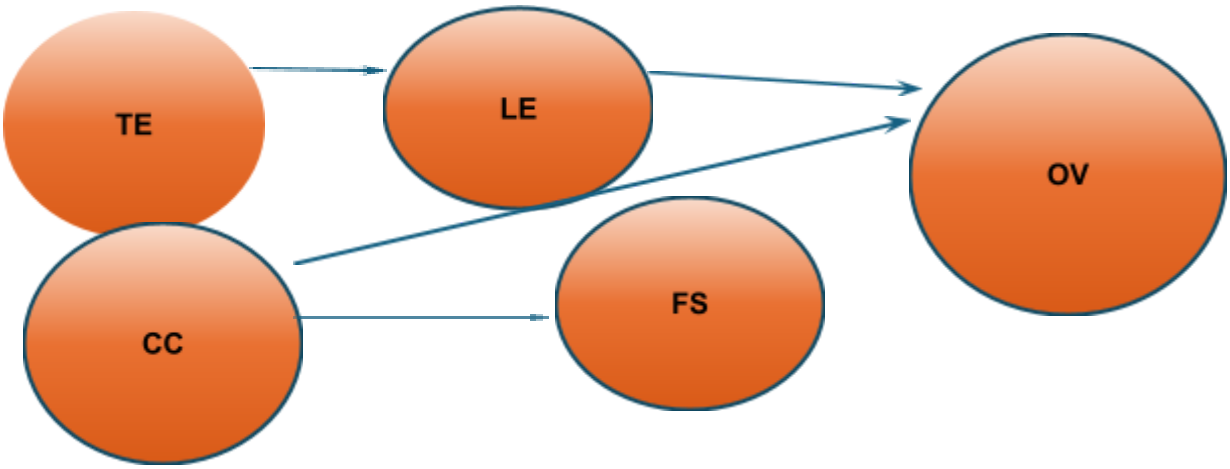
2.3 Gaps Identified

Despite the availability of these tools and technologies, several gaps remain in the domain of student feedback analysis:

1. **Under-explored Simpler Solutions:** Most existing literature emphasizes complex machine learning approaches, while simpler, tool-based solutions accessible to non-technical users are under-explored.
2. **Integration Challenges:** There is limited documentation on integrating Microsoft Copilot with Excel and Power BI for domain-specific analysis, particularly in the context of educational feedback projects.
3. **Evolving Prompt Engineering Practices:** Best practices for prompt engineering in business and education use cases are still evolving, with no standardized frameworks available.

4. **Domain-Specific Gap:** In the context of student feedback, existing solutions often focus on personalized learning rather than institutional-level feedback intelligence. This project addresses that gap by providing a framework tailored to educators and administrators.

CONCEPTUAL MODEL



TE- Teaching Effectiveness
CC- Course Content
LE-Learning Experience
FS- Feedback and Support
OV- Overall Effectiveness

This study presents a conceptual framework to examine factors influencing overall teaching effectiveness in higher education based on students' perceptions. The **independent variables** include teaching effectiveness practices, course content quality, and feedback and support. These factors influence the **mediating variable,** learning experience, which reflects students' satisfaction, motivation, engagement, and critical thinking. The **dependent variable** is overall teaching effectiveness, measured through instructor competence, ability to motivate students, and teaching satisfaction. The framework proposes that effective instruction, quality content, and strong support systems enhance students' learning experiences, which in turn improve perceptions of overall teaching effectiveness.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the overall system design, the modules used, the flow of data from input to output, and the advantages of the approach.

3.1 System Design

The system design follows a structured workflow that transforms raw student feedback into actionable insights. The process can be divided into four main stages: Data Source/Input, Data Cleaning, Analysis/Processing, and Visualization/Output.

Data Source / Input

The primary dataset consists of student feedback records collected through course evaluations and surveys. These records include both quantitative ratings (e.g., satisfaction scores) and qualitative comments (open-ended feedback). The dataset is typically stored in CSV or Excel format, containing several thousand rows of responses across multiple courses and departments.

Data Cleaning

Data cleaning is the first step to ensure accuracy and reliability. The following techniques are applied:

- Duplicate Removal: Identifying and eliminating repeated entries.
- Handling Missing Values: Filling gaps with appropriate placeholders or removing incomplete records.
- Correcting Data Types: Ensuring numerical ratings are stored as integers/floats and textual feedback as strings.
- Standardizing Formats: Normalizing date formats, correcting spelling errors, and ensuring consistent categorical labels. This pre- processing ensures that the data-set is structured and ready for analysis.

Analysis / Processing

The cleaned data-set is then analyzed using a combination of traditional and AI-assisted techniques:

- Excel Analysis: Pivot tables, formulas, and conditional formatting are used for preliminary exploration of trends.
- Power BI Measures: Advanced calculations and DAX (Data Analysis Expressions) are applied to derive metrics such as average satisfaction scores, frequency of recurring issues, and correlations between feedback themes.
- AI-Assisted Processing: Microsoft Copilot (e.g., ChatGPT, Gemini) are used to apply Natural Language Processing (NLP) and sentiment analysis. This enables categorization of feedback into themes (e.g., teaching methods, course content, assessment practices) and identification of sentiment polarity (positive, negative, neutral).

Visualization / Output

The final stage involves presenting insights in a clear and interactive format:

- Interactive Dashboards: Power BI dashboards display trends in student satisfaction, highlight strengths, and pinpoint areas needing improvement.
- Summary Reports: AI-assisted tools generate narrative summaries of feedback, providing educators with contextual understanding of student perceptions.
- Actionable Recommendations: The system outputs practical suggestions for teaching improvement, bridging the gap between raw feedback and instructional refinement.

3.2 Modules / Components Used

The system is organized into several functional modules, each responsible for a specific stage of the workflow. Together, these modules ensure that raw student feedback is transformed into actionable insights for teaching excellence.

Module 1 — Data Ingestion

Function: Loading the raw dataset into Excel or Power BI.

Details: Student feedback records (CSV/Excel format) are imported into the system. This includes both quantitative ratings and qualitative comments.

Module 2 — Data Cleaning

Function: Removing inconsistencies, duplicates, and null values.

Details: Preprocessing ensures data reliability by correcting formats, handling missing values, and standardizing categorical labels.

Module 3 — Data Analysis

Function: Applying calculations, formulas, AI prompts, or DAX measures.

Details:

- Excel pivot tables and formulas for preliminary exploration.
- Power BI measures (DAX) for advanced metrics such as satisfaction scores and recurring issue identification.
- AI-assisted analysis for Natural Language Processing (NLP) and sentiment classification.

Module 4 — Visualization / Report Generation

Function: Building dashboards, charts, or AI-generated reports.

Details: Power BI dashboards present trends in student satisfaction and teaching effectiveness. AI-generated summaries provide narrative insights for educators.

Module 5 — AI Integration

Function: Using Copilot for analysis, content generation, or insight summaries.

Details: Copilot synthesizes insights, automates reporting, and generates contextual recommendations. (e.g., ChatGPT, Gemini) assist with thematic clustering and sentiment analysis.

Module 6 — Recommendation Engine (Project-Specific)

Function: Translating analytical findings into actionable teaching strategies.

Details: Based on identified themes and sentiment trends, the system outputs practical suggestions for educators to refine instructional methods and address student concerns.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through the system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

At the highest level, the system is represented as a single process that transforms raw student feedback into actionable insights.

Textual Representation:

[Students] ---> (Raw Feedback Data: CSV/Excel) ---> [AI-Powered Feedback System] ---> (Cleaned Data, Dashboards, Reports) ---> [Educators / Administrators]

- ✔ **Input:** Raw student feedback (ratings + comments).
- ✔ **Process:** AI-powered feedback system (data cleaning, analysis, visualization).
- ✔ **Output:** Cleaned dataset, interactive dashboards, narrative reports, and recommendations.
- ✔ **External Entities:** Students (data providers) and Educators/Administrators (data consumers).

DFD Level 1 — Detailed Process Flow

This level breaks down the main process into sub-processes and data stores.

[Students] ---> (Raw Feedback Data: CSV/Excel) ---> [Sub-process 1: Data Loading & Validation]

|

v

[Data Store: Raw Dataset

[Sub-process 2: Data Cleaning & Transformation] ---> (Cleaned Dataset)

|

v

[Data Store: Structured Feedback]

[Sub-process 3: Analysis & AI Processing] ---> (Metrics, Sentiment Analysis, Thematic Clusters)

|

v

[Data Store: Analytical Results]

[Sub-process 4: Visualization & Report Generation] ---> (Dashboards, Reports, Recommendations)

|

v

[Educators / Administrators]

Sub-processes:

- ✔ **Data Loading & Validation:** Importing feedback into Excel/Power BI, checking for completeness.
- ✔ **Data Cleaning & Transformation:** Removing duplicates, handling missing values, correcting formats.

- ✔ **Analysis & AI Processing:** Applying Excel formulas, Power BI measures, and AI Copilot/Chatbot for NLP and sentiment analysis.
- ✔ **Visualization & Report Generation:** Creating dashboards, charts, and AI-generated narrative summaries.

Data Stores:

- ✔ Raw Dataset (unprocessed feedback).
- ✔ Structured Feedback (cleaned and standardized data).
- ✔ Analytical Results (metrics, sentiment scores, thematic clusters).

3.4 Advantages

The proposed methodology offers several key advantages that make it both practical and impactful in the context of student feedback analysis.

1. **Automates repetitive data cleaning tasks:** By leveraging Excel and Power BI, the system automates processes such as duplicate removal, missing value handling, and format correction, saving significant manual effort.
2. **Provides clear, visual insights:** Interactive dashboards and charts present trends in student satisfaction and teaching effectiveness in a way that is easy to understand for non-technical stakeholders such as educators and administrators.
3. **Leverages AI tools to accelerate analysis:** Microsoft Copilot and AI Chatbots assist in synthesizing insights, performing sentiment analysis, and generating narrative summaries, reducing human error and improving efficiency.
4. **Scalable across datasets and domains:** The methodology can be applied to larger datasets or adapted to other domains such as healthcare analytics, business intelligence, or employee feedback systems.
5. **Cost-effective solution:** The approach relies on widely available tools (Excel, Power BI, Copilot) that are part of the Microsoft 365 ecosystem, minimizing additional costs for institutions.

6. **Domain-specific advantage:** Unlike generic AI applications, this project is tailored specifically to educational feedback analysis, ensuring that insights are directly relevant to teaching improvement and student engagement.
7. **Timely intervention:** The system enables near real-time monitoring of feedback, allowing educators to respond quickly to emerging concerns rather than waiting until the end of a course.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4
Implementation and Results

4.1 Data Cleaning Results

Dataset Source: The dataset was collected through a **student feedback survey** designed to evaluate teaching effectiveness across different academic years **(FYBMS, SYBMS, TYBMS)** of **Shri L R Tiwari Degree College, Mumbai India.** The data was collected after sending a questionnaire through google forms, exported to **Microsoft Excel**, and later prepared for analysis in **Power BI**.

Original Dataset Size

- **Rows:** Approximately 400+ student responses
- **Columns:** 23 variables

Issues Identified

✔ Inconsistent Likert-scale responses
✔ Spelling and formatting errors
✔ Missing values in selected survey items
✔ Non-standardized categorical data
✔ Redundant and repeated entries

Cleaning Steps Applied

The following operations were performed in **Excel / Power BI**:

- Used **Find and Replace** to standardize Likert-scale responses
- Applied **text formatting and trimming** to remove extra spaces
- Corrected column names and spelling errors
- Filtered and removed incomplete or invalid records
- Validated categorical fields to ensure uniform response values
- Ensured compatibility with Power BI visualizations and measures

Cleaned Dataset Size

- **Rows:** Slightly reduced after removing incomplete entries
- **Columns:** 23 fully standardized variables

4.2 1. RAW DATASET

C11 | fx FYBMS

Timestamp	Name of the Student	Class	Faculty Name	TE1 Clarity	TE2 Interraction	TE3 Methodolc
3-24-2026 17:56:56	Avasti Shaurya	FYBMS	Prof. Saheel	Strongly Agree	Strongly Agree	Strongly Agree
3-26-2026 10:25:21	DEBNATH JIT JATINDRA	FYBMS	Prof. Saheel	Neutral	Agree	Disagree
3-26-2026 10:25:21	CHUNAWALA MOIN MOHD SALIM ALI	FYBMS	Prof. Saheel	Neutral	Disagree	Disagree
3-26-2026 10:25:21	JOSEPH JIBIN JOSEPH	FYBMS	Prof. Saheel	Neutral	Strongly Agree	Neutral
3-28-2026 19:17:18	NISHAD KAJAL RANJIT	FYBMS	Prof. Saheel	Strongly Agree	Neutral	Agree
3-28-2026 20:45:20	PANDEY NAITIK KRISHNADEV	FYBMS	Prof. Saheel	Strongly Agree	Strongly Agree	Neutral
3-28-2026 20:45:20	PANDEY PRATISHTHA SARSIJ	FYBMS	Prof. Saheel	Neutral	Strongly Agree	Strongly Agree
3-28-2026 20:54:43	PETIWALA MOHAMMAD SAAD PARVEZ	FYBMS	Prof. Saheel	Neutral	Agree	Strongly Agree
3-28-2026 20:54:43	POOJARY KAUSHIK ANANDA	FYBMS	Prof. Saheel	Neutral	Neutral	Strongly Agree
3-28-2026 20:54:43	PUROHIT HARSHITA	FYBMS	Prof. Saheel	Neutral	Neutral	Strongly Agree
3-28-2026 20:54:43	PUROHIT RAHUL KANTILAL	FYBMS	Prof. Saheel	Agree	Strongly Agree	Strongly Agree
3-28-2026 21:58:23	RAJPUT KASHISH AJIT	FYBMS	Prof. Saheel	Agree	Strongly Agree	Strongly Agree
3-28-2026 21:59:23	RAJPUT SAKSHI VICKY	FYBMS	Prof. Saheel	Agree	Strongly Agree	Strongly Agree
3-28-2026 21:59:56	RANA KRISMA PRAKASH	FYBMS	Prof. Saheel	Agree	Strongly Agree	Strongly Agree
3-28-2026 21:59:56	RATHOD ISHA RAHUL	FYBMS	Prof. Saheel	Strongly Agree	Neutral	Strongly Agree
3-28-2026 21:59:56	RATHOD NAYAN SHAILESHBHAI	FYBMS	Prof. Saheel	Strongly Agree	Neutral	Strongly Agree
3-28-2026 22:33:55	RATHOD TANUSHRI MUKESHKUMAR	FYBMS	Prof. Saheel	Strongly Agree	Neutral	Neutral
3-28-2026 22:45:12	RAY HARSH ANIL KUAMR SHARMA	FYBMS	Prof. Saheel	Strongly Agree	Neutral	Strongly Agree

RAW | Data Cleaning and Duplicaton

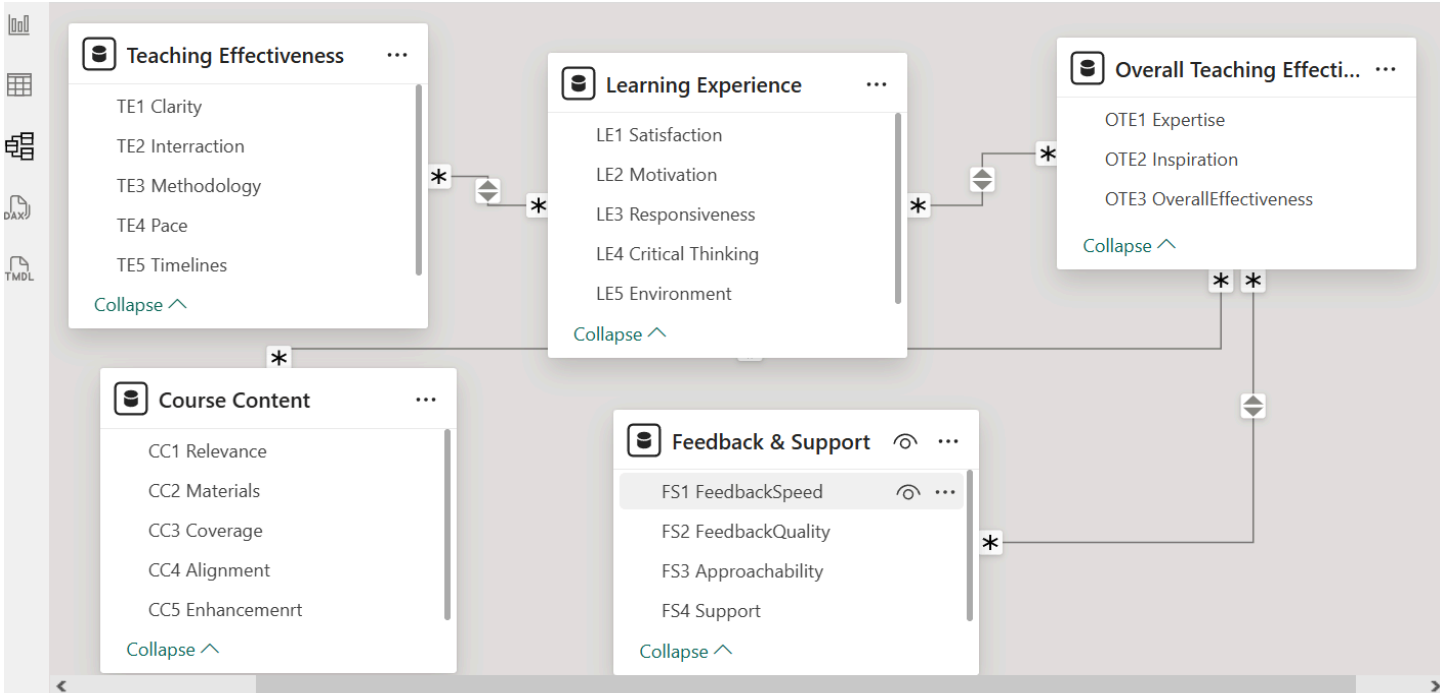
Activate Windows
Go to Settings to activate Windows.

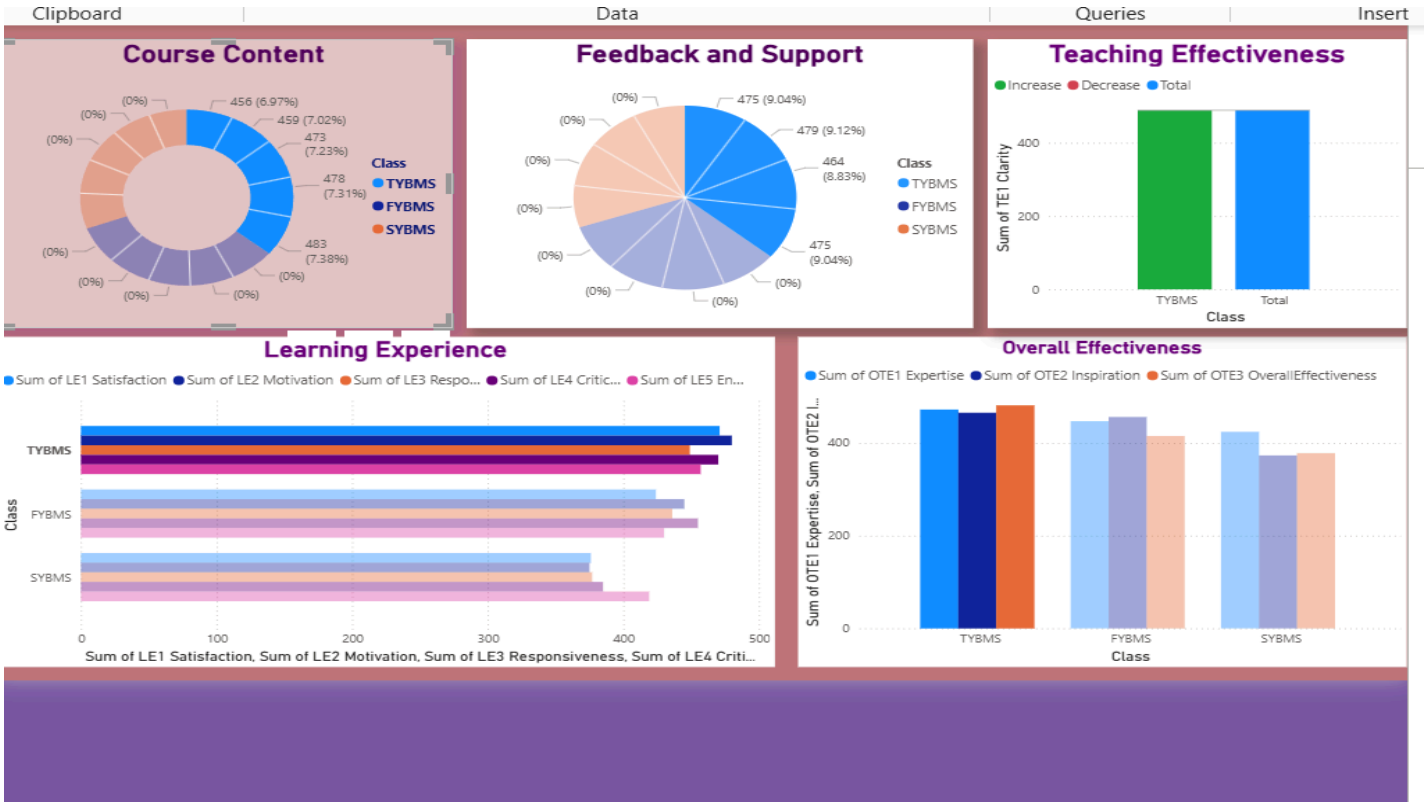
Class	TE1 Clarity	TE2 Interraction	TE3 Methodology	TE4 Pace	TE5 Timelines	CC1 Relevance	CC2 Materials	CC3 Coverage	CC4 Alignmer
FYBMS	5	5	5	5	5	5	5	5	
FYBMS	4	5	5	4	5	5	3	5	
FYBMS	4	5	5	2	5	5	4	5	
FYBMS	4	5	5	2	5	5	4	5	
FYBMS	4	5	5	1	5	5	4	5	
FYBMS	5	5	5	5	4	4	5	5	
FYBMS	5	5	5	4	4	3	2	5	
FYBMS	5	5	5	4	3	2	2	5	
FYBMS	3	5	4	4	3	4	2	5	
FYBMS	3	5	4	4	3	2	2	5	
FYBMS	2	5	4	4	3	5	5	4	
FYBMS	2	5	4	4	3	5	5	4	
FYBMS	2	5	4	3	3	4	5	4	
FYBMS	2	5	3	3	3	4	5	4	
FYBMS	4	5	3	3	5	4	5	4	
FYBMS	4	5	3	3	5	4	5	4	
FYBMS	4	5	5	3	5	4	5	4	
FYBMS	4	5	4	3	5	4	5	3	
FYBMS	4	5	3	3	5	4	5	3	
FYBMS	4	5	3	3	5	4	5	3	

Table: Table 1 (Dataset-Coding) (309 rows)

4.3 Visualization / Output Results

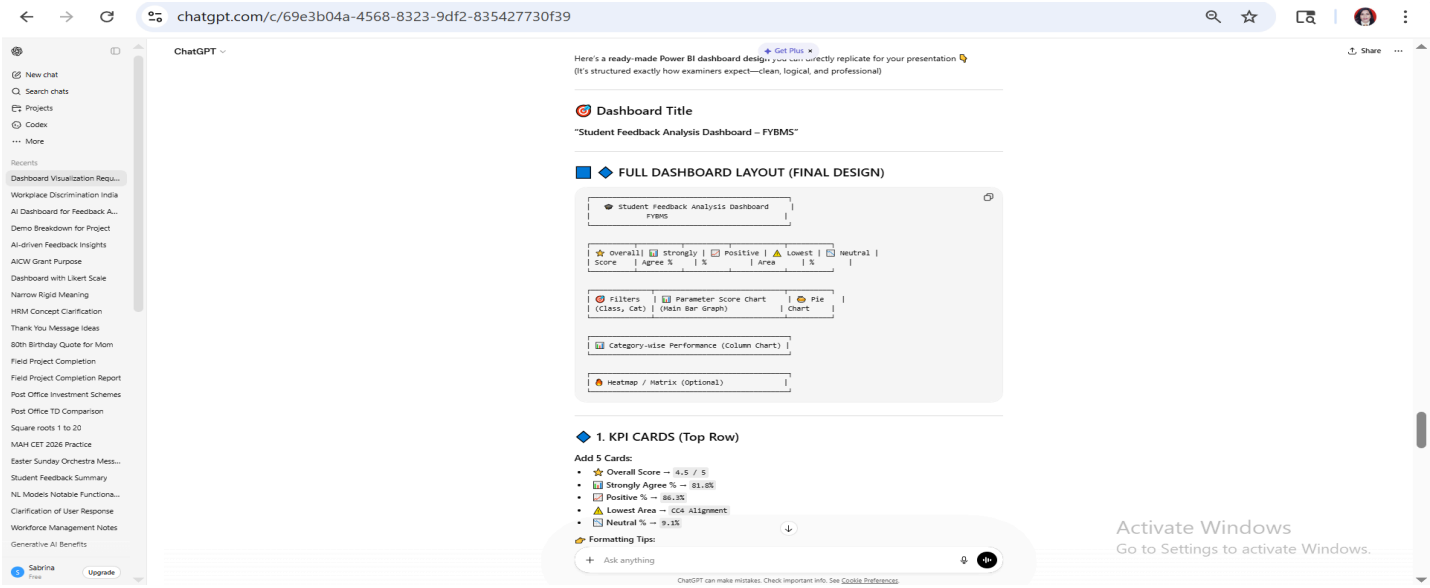
1. Data Model





4.3 AI Tool Integration Results

ChatGPT



Prompt 1: Explain how data is cleaned, transformed, and analyzed using your tools.] So the data was split with the help of Power BI tool and then the null columns were deleted and it was first transformed and then it was like it showed the null columns which was deleted and then taken now for analysis.

Prompt 2: Give me exact Power BI DAX formulas.

Prompt 3: what are the KPIs for my project?

Output Quality:

The AI-generated output was accurate, well-structured, and easy to understand. It provided clear insights, meaningful visualizations, and properly formatted data, which made the analysis more professional and reliable.

How it Enhanced the Project:

ChatGPT helped speed up the process by generating datasets, Power BI layouts, and DAX formulas instantly. It improved the quality of analysis by identifying key insights, KPIs, and weak areas, making the project more effective and data-driven.

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- After data cleaning and transformation, 309 valid responses were analyzed, ensuring accuracy and consistency in the dashboard insights.
- The overall average satisfaction score was 4.17 out of 5, indicating a high level of student satisfaction with teaching and learning practices.
- Overall Teaching Effectiveness (4.20) and Teaching Methodology (4.36) emerged as the strongest areas, showing that students highly value clarity and effective teaching methods.
- Feedback & Support-related aspects, such as student satisfaction and instructor approachability (scores around 3.9–4.0), were comparatively lower, indicating areas for improvement.
- Approximately 76.4% of responses fell under Agree and Strongly Agree, reflecting a predominantly positive perception among students.
- Class-wise analysis showed that SYBMS and TYBMS students reported slightly higher satisfaction compared to FYBMS, suggesting improved learning experiences at advanced levels.
- Question-level analysis helped identify specific strengths (e.g., teaching methods, feedback quality) and weak areas (e.g., engagement, support systems), enabling targeted improvements.
- The use of Power BI enabled efficient data modeling, visualization, and automated insight generation, simulating an AI-powered feedback analysis system.

5.2 Project Links

Share the relevant links to your project work for evaluation.

1. https://github.com/drsabrinadsouza-crypto/AI-powered-Student-feedback-system

2. https://drive.google.com/file/d/1Q0K9GZ4mdD5FVhTrLB_AE_16a3Tuj4Pv/view?usp=drive_link

5.3 Limitations

- ★ The study is based on responses from 309 students of a single institution, which may limit the generalizability of the findings to other colleges or educational contexts.

- ★ The analysis relies on self-reported data, which may be subject to bias, as students might provide socially desirable responses rather than completely objective feedback.
- ★ The dataset is based on a Likert scale, which captures perception but may not fully reflect the depth of student experiences or qualitative insights.
- ★ The dashboard provides descriptive analytics only and does not include advanced predictive or real-time AI capabilities due to tool and scope limitations.
- ★ The data was collected at a single point in time, so it does not capture changes in student perception over a longer period.
- ★ Some aspects such as emotional engagement, teaching style nuances, and individual student differences cannot be fully captured through structured questionnaire data.
- ★ The project depends on manual data preparation steps (cleaning, merging, unpivoting), which may introduce minor human errors if not handled carefully.

5.4 Future Work

- ➔ Integrate real-time data pipelines using tools like Power Automate or API connections, enabling continuous collection and live updating of student feedback dashboards.
- ➔ Apply machine learning techniques such as sentiment analysis, predictive modeling, and trend forecasting to move from descriptive insights to predictive and prescriptive analytics.
- ➔ Develop a web-based or Power Apps interface to make the dashboard easily accessible to faculty and administrators without requiring technical expertise.
- ➔ Expand the dataset by including multiple institutions, courses, and semesters, allowing for more comprehensive and comparative analysis across different educational environments.
- ➔ Incorporate qualitative feedback (open-ended responses) and use Natural Language Processing (NLP) to extract deeper insights into student opinions and experiences.
- ➔ Enhance the system into a fully AI-powered recommendation engine that suggests specific teaching improvements based on identified weak areas.
- ➔ Implement role-based dashboards for different stakeholders (faculty, management, students) to provide customized insights and decision support.
- ➔ Develop a standardized AI prompt library for tools like Copilot or chatbots to automate report generation, insights extraction, and academic decision-making.

- ➔ Introduce longitudinal analysis by collecting feedback over multiple semesters to track trends and measure the impact of improvements over time.

5.5 Conclusion

This project, titled ***"AI-Powered Student Feedback Intelligence: Transforming Insights into Teaching Excellence,"*** successfully analyzed student feedback collected from 309 respondents across FYBMS, SYBMS, and TYBMS using a structured Likert-scale questionnaire. The primary objective was to transform raw feedback data into meaningful insights that support continuous improvement in teaching and learning practices.

Through the use of Power BI for data cleaning, transformation, and visualization, the dataset was systematically modeled and analyzed across five key dimensions: Teaching Effectiveness, Course Content, Learning Experience, Feedback & Support, and Overall Teaching Effectiveness. The results revealed an overall high satisfaction level with an average score of 4.17, indicating that students generally perceive the quality of teaching positively.

The analysis identified Overall Teaching Effectiveness and Teaching Methodology as key strengths, while areas such as student satisfaction, engagement, and support mechanisms showed relatively lower scores, indicating opportunities for enhancement. Additionally, class-wise comparison highlighted that senior students (SYBMS and TYBMS) reported slightly higher satisfaction levels, suggesting improved learning experiences over time.

All the objectives of the project were successfully achieved, including data collection, processing, visualization, and insight generation. The project demonstrates how data-driven tools can be leveraged to simulate AI-powered feedback intelligence, enabling institutions to identify patterns, monitor performance, and make informed academic decisions.

Overall, this project adds significant value to the domain of education by providing a scalable and analytical framework for evaluating teaching effectiveness, ultimately contributing to improved student outcomes and institutional excellence.

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APPENDICES

Appendix A — Sample Dataset

C11 | fx | FYBMS

Timestamp	Name of the Student	Class	Faculty Name	TE1 Clarity	TE2 Interraction	TE3 Methodolc
3-24-2026 17:56:56	Avasti Shaurya	FYBMS	Prof. Saheel	Strongly Agree	Strongly Agree	Strongly Agree
3-26-2026 10:25:21	DEBNATH JIT JATINDRA	FYBMS	Prof. Saheel	Neutral	Agree	Disagree
3-26-2026 10:25:21	CHUNAWALA MOIN MOHD SALIM ALI	FYBMS	Prof. Saheel	Neutral	Disagree	Disagree
3-26-2026 10:25:21	JOSEPH JIBIN JOSEPH	FYBMS	Prof. Saheel	Neutral	Strongly Agree	Neutral
3-28-2026 19:17:18	NISHAD KAJAL RANJIT	FYBMS	Prof. Saheel	Strongly Agree	Neutral	Agree
3-28-2026 20:45:20	PANDEY NAITIK KRISHNADEV	FYBMS	Prof. Saheel	Strongly Agree	Strongly Agree	Neutral
3-28-2026 20:45:20	PANDEY PRATISHTHA SARSIJ	FYBMS	Prof. Saheel	Neutral	Strongly Agree	Strongly Agree
3-28-2026 20:54:43	PETIWALA MOHAMMAD SAAD PARVEZ	FYBMS	Prof. Saheel	Neutral	Agree	Strongly Agree
3-28-2026 20:54:43	POOJARY KAUSHIK ANANDA	FYBMS	Prof. Saheel	Neutral	Neutral	Strongly Agree
3-28-2026 20:54:43	PUROHIT HARSHITA	FYBMS	Prof. Saheel	Neutral	Neutral	Strongly Agree
3-28-2026 20:54:43	PUROHIT RAHUL KANTILAL	FYBMS	Prof. Saheel	Agree	Strongly Agree	Strongly Agree
3-28-2026 21:58:23	RAJPUT KASHISH AJIT	FYBMS	Prof. Saheel	Agree	Strongly Agree	Strongly Agree
3-28-2026 21:59:23	RAJPUT SAKSHI VICKY	FYBMS	Prof. Saheel	Agree	Strongly Agree	Strongly Agree
3-28-2026 21:59:56	RANA KRISMA PRAKASH	FYBMS	Prof. Saheel	Agree	Strongly Agree	Strongly Agree
3-28-2026 21:59:56	RATHOD ISHA RAHUL	FYBMS	Prof. Saheel	Strongly Agree	Neutral	Strongly Agree
3-28-2026 21:59:56	RATHOD NAYAN SHAILESHBHAI	FYBMS	Prof. Saheel	Strongly Agree	Neutral	Strongly Agree
3-28-2026 22:33:55	RATHOD TANUSHRI MUKESHKUMAR	FYBMS	Prof. Saheel	Strongly Agree	Neutral	Neutral
3-28-2026 22:45:12	RAY HARSH ANIL KUAMR SHARMA	FYBMS	Prof. Saheel	Strongly Agree	Neutral	Strongly Agree

RAW | Data Cleaning and Duplicaton | +

Activate Windows
Go to Settings to activate Windows.

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Appendix B — Key Excel Formulas / DAX Measures Used

Tool	Formula / Measure	Purpose / Description
Excel	=IFERROR(VLOOKUP(...), "Not Found")	Used for data validation and handling missing or unmatched values
Excel	=COUNTIF(range, criteria)	Used to identify and count duplicate or repeated entries
Power BI (DAX)	Total Responses = COUNT('Table'[Score])	Calculates the total number of responses
Power BI (DAX)	Average Score = AVERAGE('Table'[Score])	Computes overall and section-wise average ratings
Power BI (DAX)	Positive % = DIVIDE(CALCULATE(COUNT('Table'[Score]), 'Table'[Score] >= 4), COUNT('Table'[Score]))	Calculates percentage of positive responses (Agree & Strongly Agree)

Tool	Formula / Measure	Purpose / Description
Power BI (DAX)	Negative % = DIVIDE(CALCULATE(COUNT('Table'[Score]), 'Table'[Score] <= 2), COUNT('Table'[Score]))	Calculates percentage of negative responses
Power BI (DAX)	Response Type = SWITCH(TRUE(), Score=1,"Strongly Disagree", Score=2,"Disagree", Score=3,"Neutral", Score=4,"Agree", Score=5,"Strongly Agree")	Categorizes Likert-scale responses for visualization
Power BI (DAX)	Satisfaction Index = AVERAGE('Table'[Score])	Represents overall performance indicator
Power BI	Unpivot Columns	Converts wide-format data into long format for better analysis
Power BI	Merge & Append Queries	Combines multiple sheets into a unified dataset

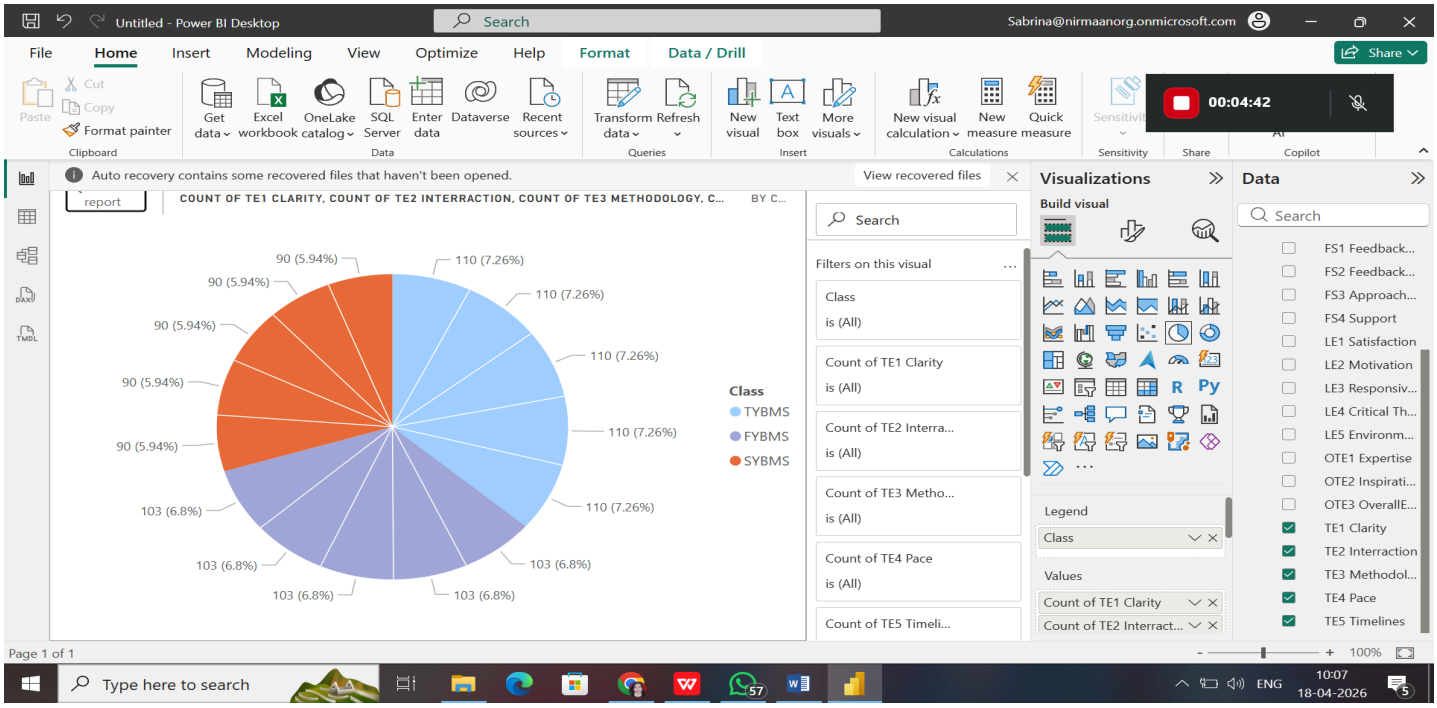
Appendix C — AI Prompts Used

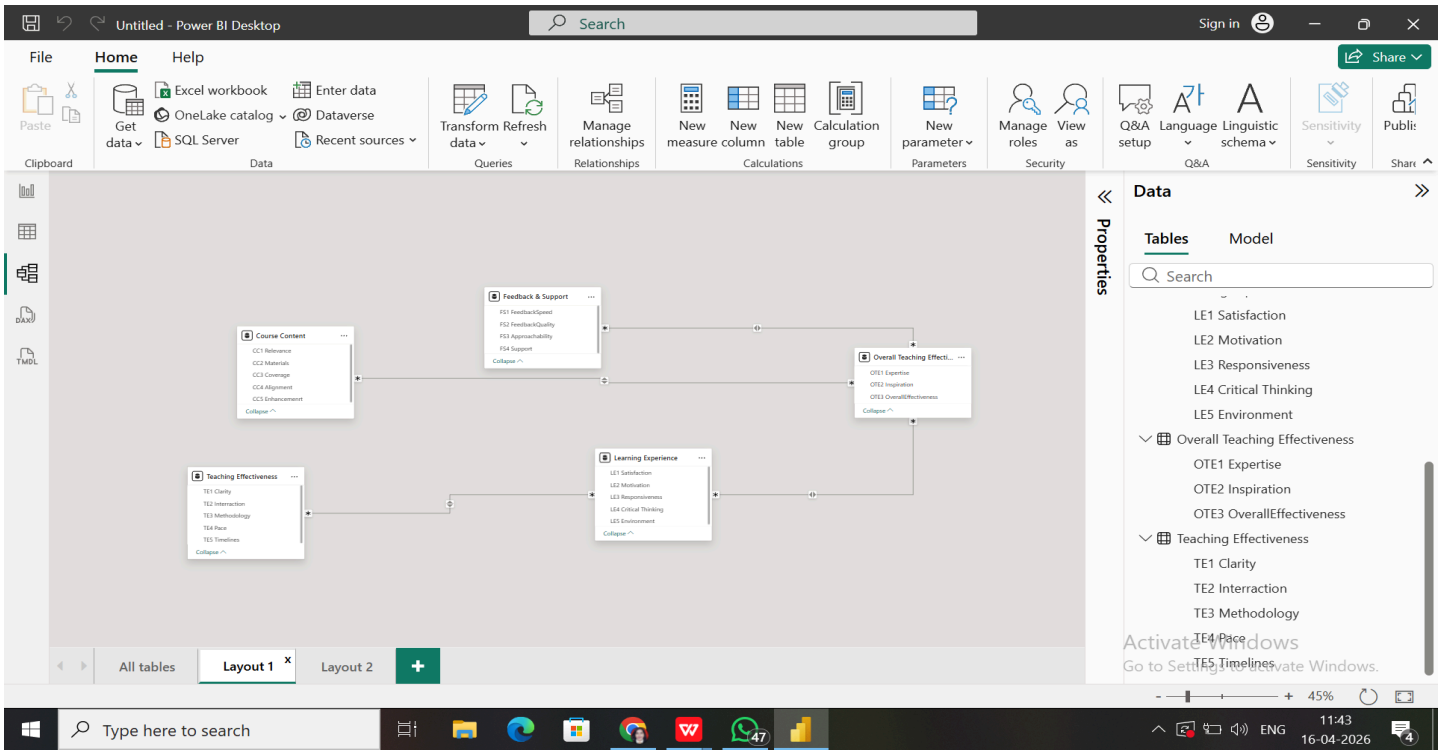
Prompt 1: Explain how data is cleaned, transformed, and analyzed using your tools.] So the data was split with the help of Power BI tool and then the null columns were deleted and it was first transformed and then it was like it showed the null columns which was deleted and then taken now for analysis.

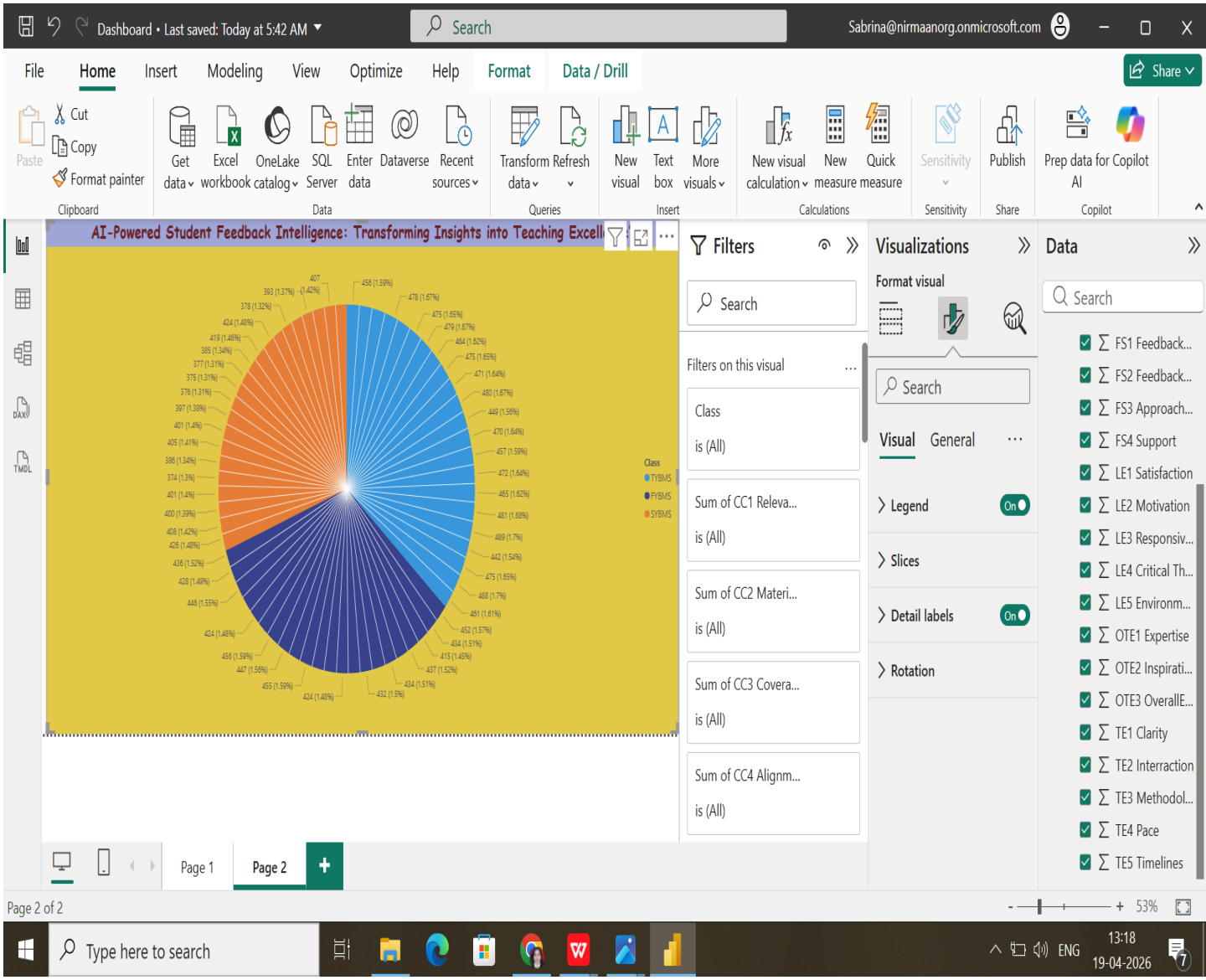
Prompt 2: Give me exact Power BI DAX formulas.

Prompt 3: what are the KPIs for my project?

Appendix D — Screenshot Gallery



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File
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Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Share
Cut
Copy
Paste
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for Copilot AI
Copilot
AI-Powered Student Feedback Intelligence: Transforming Insights into Teaching Excell
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Detail labels
On
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FS2 Feedback...
FS3 Approach...
FS4 Support
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LE2 Motivation
LE3 Responsiv...
LE4 Critical Th...
LE5 Environm...
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OTE2 Inspirati...
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